RESHMA MERIN THOMAS

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Summary

Master's student in Computer Science specializing in machine learning and computer vision, with experience building 2D and 3D perception models using Python and PyTorch. Skilled in object detection, model optimization, and data-driven problem solving, with a strong interest in developing efficient and scalable AI solutions.

Education

SRM Institute of Science and Technology
B.Tech in Computer Science and Engineering
spz in Big Data Analytics
Jul 2020 – Jun 2024
GPA: 3.5 / 4.0

California State University, Sacramento
M.S. in Computer Science
Aug 2024 – Jul 2026 (Expected)
GPA: 3.1 / 4.0

Experience

Data Analyst, Information Resources & Technology, CSUS – CA — Aug'25 - Present

- Developed automation scripts and performed data validation using Python and SQL best practices to streamline technical workflows, improve data accuracy, and support accessibility remediation efforts across multiple digital platforms.
- Collaborated with cross-functional teams to document processes, analyze web pages, and enhance compliance with data privacy and accessibility standards, leveraging tools such as Canvas Ally, LMS-integrated applications, and other web-based systems.

Project Trainee, Center of AI and Robotics, DRDO – Bangalore, India — Jan'24 - Mar'24

- Collaborated with the Intelligent Surveillance and Robotics Division (ISRD) to enhance perception systems for autonomous and electric vehicles, focusing on advanced sensor integration and spatial understanding. Researched and implemented Neural Radiance Fields (NeRF) for high-fidelity volumetric 3D scene reconstruction from sparse image datasets, contributing to ongoing full-stack perception and simulation research.
- Leveraged NVIDIA's Instant-NGP framework for real-time neural rendering and accelerated ray tracing, optimizing computational efficiency and achieving a 23% performance improvement in depth completion, 3D semantic segmentation, and free-space estimation through NeRF-based sensor fusion techniques.

Projects

Football Commentary System using YOLOv8 and ByteTrack — <github link>

- Built a real-time football commentary system using YOLOv8 and ByteTrack for detection/tracking, with event recognition and TTS for automated game narration.

Perception of Autonmous Vehilces using Neural Radiance Field(NeRF) — 2024

- Built a NeRF-based perception system for autonomous vehicles to reconstruct 3D scenes, improving depth estimation, object recognition, and navigation.

P2P File Sharing using Etherium and Blockchain — <github link>

- Developed a P2P file sharing system with blockchain-based logging using Docker, Flask, and Web3.py for transparent, tamper-proof file transfer tracking.

Extra Curricular

Volunteer - IT support, The Episcopal Diocese of North California - CA — Oct'24 - Dec'24

- Provided IT and technical support in softwares, managing systems, data, and processes to streamline daily operations.

Skills

- **Programming & Development:** Proficient in Python, C++, MATLAB, R, JavaScript, Angular, and SQL (MySQL, MongoDB); experienced with AWS, Docker, and modern software engineering practices.
- **Machine Learning & Data Science:** Skilled in Deep Learning (CNNs, RNNs, Transformers), Computer Vision, NLP, Data Analysis, Feature Engineering, and Statistical Modeling.
- **Core Strengths:** Strong analytical thinking, problem solving, research orientation, and rapid learning ability.

Publication

- Mahajan K., Thomas R., Patel S., Thupally A. and Molokwu B. (2025). One-Shot Learning, Video-to-Audio Commentary System for Football and/or Soccer Games. In Proceedings of the 21st International Conference on Web Information Systems and Technologies - Volume 1: WEBIST; ISBN 978-989-758-772-6, SciTePress, pages 335-342. DOI: 10.5220/0013692000003985